

DevOps Mastery Program

Training Program

Technology: DevOps | Duration: 10 day(s) | Mode: Online | Trainer: DevOps Trainer

Program Roadmap

Day 1: DevOps Foundations | Tools: Git + GitHub + Docker | Jira: Set up initial sprint board

Day 2: Linux for DevOps | Tools: Linux + Bash | Jira: Track server configuration tasks

Day 3: Containerization | Tools: Docker + Docker Compose | Jira: Create containerization tasks

Day 4: Kubernetes Essentials | Tools: Kubernetes + kubectl + Helm | Jira: Plan cluster deployment work

Day 5: Cloud Infrastructure | Tools: AWS + Azure + GCP | Jira: Manage cloud infrastructure tickets

Day 6: Infrastructure as Code | Tools: Terraform + Pulumi | Jira: Automate infrastructure changes

Day 7: CI/CD Pipelines | Tools: Jenkins + GitHub Actions + GitLab CI | Jira: Automate release pipelines

Day 8: Monitoring & Observability | Tools: Prometheus + Grafana + ELK | Jira: Create monitoring tasks

Day 9: Security & Compliance | Tools: Vault + SonarQube + Snyk | Jira: Track security remediation

Day 10: Release & Support | Tools: Jira + Slack + PagerDuty | Jira: Plan release and support handoff

Day 1: DevOps Foundations

Tools: Git + GitHub + Docker | Jira Focus: Set up initial sprint board

Topic Coverage

CI/CD concepts

Infrastructure as Code

DevOps toolchain overview

Version control with Git

Collaboration workflows

Lab Task

- Configure Git repo and CI pipeline

Jira Practice

- Set up initial sprint board

Day 2: Linux for DevOps

Tools: Linux + Bash | Jira Focus: Track server configuration tasks

Topic Coverage

Linux shell basics

File system permissions

Process management

Networking commands

Bash scripting

Lab Task

- Deploy a Linux VM and automate health checks

Jira Practice

- Track server configuration tasks

Day 3: Containerization

Tools: Docker + Docker Compose | Jira Focus: Create containerization tasks

Topic Coverage

Docker architecture

Container lifecycle

Image builds

Docker Compose

Best practices

Lab Task

- Build and run a microservice stack in Docker

Jira Practice

- Create containerization tasks

Day 4: Kubernetes Essentials

Tools: Kubernetes + kubectl + Helm | Jira Focus: Plan cluster deployment work

Topic Coverage

K8s architecture

Pods and deployments

Services and networking

ConfigMaps and Secrets

Helm basics

Lab Task

- Deploy an app to a Kubernetes cluster

Jira Practice

- Plan cluster deployment work

Day 5: Cloud Infrastructure

Tools: AWS + Azure + GCP | Jira Focus: Manage cloud infrastructure tickets

Topic Coverage

AWS basics

Azure fundamentals

GCP overview

VPC and networking

Storage services

Lab Task

- Provision cloud resources using IaC

Jira Practice

- Manage cloud infrastructure tickets

Day 6: Infrastructure as Code

Tools: Terraform + Pulumi | Jira Focus: Automate infrastructure changes

Topic Coverage

Terraform core

State management

Modules

Provisioning workflows

Policy as code

Lab Task

- Write Terraform for a multi-tier app

Jira Practice

- Automate infrastructure changes

Day 7: CI/CD Pipelines

Tools: Jenkins + GitHub Actions + GitLab CI | Jira Focus: Automate release pipelines

Topic Coverage

Pipeline design

Build automation

Testing integration

Deployment strategies

Rollback plans

Lab Task

- Build a CI/CD pipeline for code deployment

Jira Practice

- Automate release pipelines

Day 8: Monitoring & Observability

Tools: Prometheus + Grafana + ELK | Jira Focus: Create monitoring tasks

Topic Coverage

Logging

Metrics

Tracing

Alerting

Dashboarding

Lab Task

- Set up observability for a sample app

Jira Practice

- Create monitoring tasks

Day 9: Security & Compliance

Tools: Vault + SonarQube + Snyk | Jira Focus: Track security remediation

Topic Coverage

Secure CI/CD

IAM best practices

Secret management

Vulnerability scanning

Compliance checks

Lab Task

- Integrate security scanning into pipeline

Jira Practice

- Track security remediation

Day 10: Release & Support

Tools: Jira + Slack + PagerDuty | Jira Focus: Plan release and support handoff

Topic Coverage

Release management

Incident response

Post-mortem practices

Continuous improvement

Cost optimization

Lab Task

- Create an incident response runbook

Jira Practice

- Plan release and support handoff

Tools & Software

- Git
- Docker
- Kubernetes
- Terraform
- AWS
- Azure
- GCP
- Jenkins
- GitHub Actions
- Prometheus
- Grafana
- ELK
- Vault
- SonarQube
- Snyk
- Linux
- Bash
- Helm
- Pulumi
- PagerDuty

Certification Roadmap

- AWS Certified DevOps Engineer
- Google Professional DevOps Engineer
- Microsoft Certified: DevOps Engineer Expert
- Certified Kubernetes Administrator